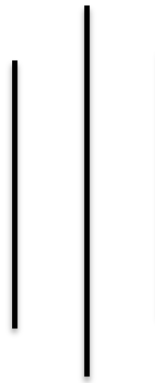


**TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING (IOE)
WESTERN REGIONAL CAMPUS
DEPARTMENT OF CIVIL ENGINEERING
Lamachaur-16, Pokhara**



**A Report on
Field Excursion to Bijayapur-1 Hydropower Plant**

Prepared by:

Sonika Dhungana

(Roll no- PAS078BCE129)

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Lastly, my appreciation goes to all my friends and fellow participants whose cooperation and companionship made the excursion both productive and enjoyable.

PREFACE

This report on the **Bijaypur-I Hydropower Plant** documents the observations and key learnings from a one-day field excursion conducted on 24 Shrawan 2082, as part of the Hydropower Engineering syllabus for the Bachelor of Civil Engineering (Part IV/I) program at the Institute of Engineering (IOE), Western Regional Campus. The visit aimed to provide students with practical exposure to hydropower generation processes and their significance in energy sector.

The objective of this study is to present an overview of the Bijaypur-I Hydropower Project, including its purpose, operational scope, and area of influence. The report also highlights the project's main features, outlines existing challenges, and suggests possible improvements.

Field excursions are an essential part of engineering education, enabling students to bridge the gap between classroom learning and real-world application. This report seeks to enhance understanding of hydropower systems and inspire greater appreciation for their contribution to Nepal's socio-economic growth and environmental sustainability.

Sonika Dhungana

PAS078BCE129

Fourth-Year Civil Engineering Student

IOE Western Regional Campus

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1. Introduction

1.1 Background

Hydropower is the generation of electricity by harnessing the energy of flowing water. Water stored at a higher elevation possesses potential energy, which is converted to kinetic energy as it flows down the penstock. Through a nozzle, it strikes the turbine blades, producing rotational energy. The turbine shaft, coupled with a generator, then converts this into electrical energy.

Hence, Hydropower is the conversion of potential energy into usable electrical energy through successive stages of energy transformation. Thanks to the natural hydrologic cycle, the flow of water is sustained, making hydropower the most widely used form of renewable electricity worldwide.

Based on the method of generation, hydropower projects are classified into:

- Storage Type Project – water stored in a reservoir is released to generate electricity.
- Pumped Storage Type Project – water is pumped from a lower to an upper reservoir for later generation.
- Run-of-River (RoR) Project – utilizes a river's natural flow without significant storage.

1.2 Status of Hydropower in Nepal

Nepal's mountainous terrain and abundant river systems give it one of the highest hydropower potentials in the world about 103 GW as estimated by Bajracharya (2015), with roughly 42 GW considered economically feasible. Several large and medium-scale projects are in operation or under construction, with increasing participation from private Independent Power Producers (IPPs). As of mid-2025, the country's total installed electricity capacity stands at approximately **3,602 MW**, while slightly more recent reports indicate a figure of **3,878 MW**. The majority of this capacity, roughly 95 % originates from hydropower. Government policies also promote micro- and small-hydro schemes to electrify rural communities.

The proposed Karnali Chisapani Storage Hydropower Project (10,800 MW) will be the country's largest once completed. Other notable projects include Upper Tamakoshi (456 MW), Arun III (900 MW), and Upper Karnali (900 MW).

While Nepal often produces surplus electricity in the wet season, dry-season output declines sharply due to the dominance of run-of-river (RoR) plants with minimal storage, leading to seasonal imports from India. Transmission and distribution losses remain high, at nearly 30% of total generation.

Hydropower remains central to Nepal's energy strategy, with potential to enhance energy security, support economic growth, and expand cross-border electricity trade if challenges in financing, infrastructure, environment, and governance are effectively addressed.

1.3 Field Visit Background and Rationale

In our one-day field visit, we sought to observe a run-of-river (ROR) project located near our vicinity to gain a practical understanding of its components and operation. This provided us with a unique opportunity to see the theoretical concepts we've studied applied in a real-world setting. The Bijayapur-I Hydropower Project served as the ideal subject for our study, allowing us to connect our academic knowledge directly to a tangible and functional power generation facility.

1.4 Methodology

The methodology used for data collection was comprehensive. We gathered information through direct observation and photographic documentation of the project's components. Additionally, we relied on an on-site explanation provided by our supervising professor and held informal discussions within the site. To supplement this on-site data, further research was conducted using internet sources and past reports to provide a more complete and well-rounded analysis.

1.5 Objectives

The primary objective of our field visit was to bridge the gap between our theoretical knowledge and a practical, real-world application by conducting a comprehensive study of the Bijayapur-I Hydropower Project.

Our secondary objectives were:

- To identify and understand the various components of a run-of-river hydropower project, from the intake to the powerhouse.
- To observe the layout of the project's components and analyze their specific working mechanisms.
- To comprehend the entire process of power generation, particularly within the powerhouse.
- To assess the current operational condition of the hydropower facility.

2. Bijayapur-1 Hydropower

2.1 Introduction

The Bijaypur-I Small Hydropower Project is a Run-of-River (RoR) type plant located in Pokhara-Lekhnath-26, Kaski, Nepal. Situated approximately 1.5 km west of Kalika Chowk, the project harnesses the flow of the Bijaypur Khola and surplus water from the Seti River's irrigation system. It was commissioned on Bhadra 05, 2069 BS (August 21, 2012) and is privately owned, developed, and operated by Bhagawati Hydropower Development Company Pvt. Ltd., an Independent Power Producer (IPP). The project, with a gross head of 68 meters and a maximum installed capacity of 4.5 MW, generates approximately 29.33 GWh of energy annually.

The total project cost was approximately NRs. 80 crores, financed by a consortium of banks including Rastriya Banijya Bank, Nabil Bank, and Nepal Bank. The project holds a 30-year generation license, after which ownership will be transferred to the Government of Nepal. A key feature of this development is its cascade design: the tailrace water from the Bijaypur-I project is utilized by the downstream Bijaypur-II Hydropower Project, optimizing the use of the Bijaypur Khola's water resources. The electricity generated is transmitted through a 33kV transmission line.

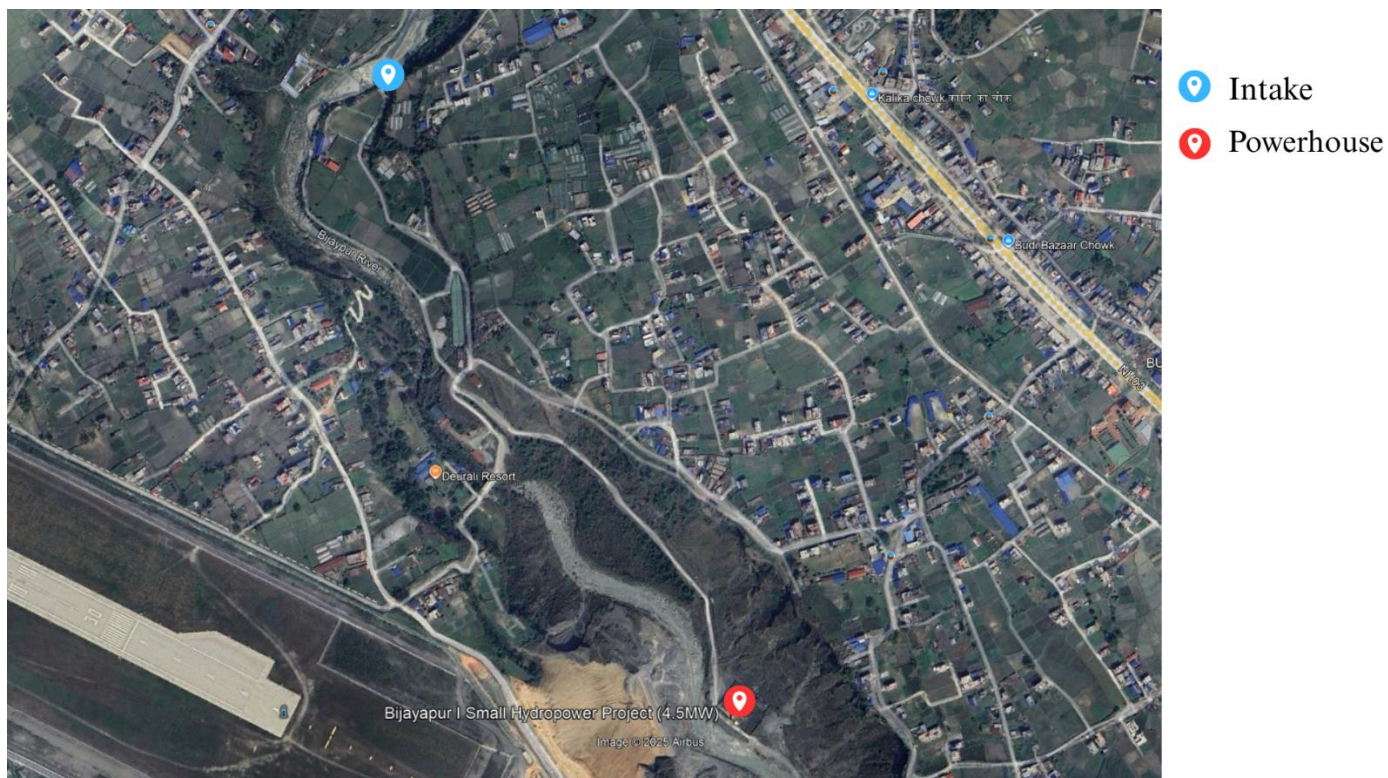


Figure 1. Location of Bijayapur I Small Hydropower Project

2.2 Salient Features

Parameter	Description
Developer	Bhagawati Hydropower Development Company (p) Ltd
Project	Bijaypur-I Small Hydropower Project
District	Kaski
Location	Pokhara, Lekhnath Municipality
Water Source	Bijaypur Khola + Irrigation Drainage from Seti River
Scheme Type	Run-off-River (Medium head plant)
Access Road	2 km
Installed Capacity	4500 kW (4.5 MW)
Annual Design Generation	29.33 GWh
River Discharge	10-20 m ³ /s
Design Discharge	8.3 m ³ /s
Gross Head	68.0 m
Rated Design Head	65.4 m
Catchment Area at Intake	65.3 sq km
Probability of Exceedance	45 %
Headworks	Stone lined free flow weir 30m
Weir Crest Elevation	EL 752 m
Under sluice	2 vertical sluice gates
Intake Type	Side intake with coarse rack
Connecting Channel	60m RCC Rectangular closed canal
Gravel Trap	RCC Rectangular (7 × 3.5m)
Headrace Canal Length	242m RCC closed canal

Desander	RCC Dufour Type $\times$ 2 basins with forebay at the end
Daily Regulating Pond Capacity	3000 m ³
Penstock Pipe	Welded Ribbed Pipe 2.5m diameter, 2 nos. with 4.15 m ³ /s discharge in each
Pipe Material	Mild steel (IS 2062-B)
Penstock Length	634 m
Number of Supports	206 nos.
Number of Anchor Blocks	10 nos. including Y and Bellmouth block
Powerhouse	Surface (23m $\times$ 11m) with outdoor switchyard
Tailrace	Cut and Cover RCC, 302m
Turbine	Horizontal shaft Francis, 2 units
Transmission	4 km, 33kV Single Circuit
Interconnection/Metering Point	NEA 132 kV at Lekhnath Substation
Construction Started	2067 Kartik
Commissioned Date	Bhadra 05, 2069
Rate per Unit of Energy	Rs 4.80 for wet season and 8.40 for dry season
Project Cost	71 crore (without IDC)
Total Cost at Commissioning	80 crore
Financed by	Rastriya Banijya Bank, Nabil Bank & Nepal Bank Ltd.

2.3 Components of Bijaypur-1 Hydropower Project

The components of ROR Hydropower Scheme that we saw in the Field Excursion are as follows:

1. Weir
2. Undersluice
3. Intake
4. Intake canal/culvert
5. Gravel trap
6. Spillway
7. Settling Basin
8. Fore bay
9. Penstock
10. Powerhouse
11. Tailrace canal
12. Transmission
13. Access Road

1. Weir

A weir is a small overflow type barrier built across a river, stream, or canal to control the flow of water.

A 30 m masonry gravity structured stone-lined free flow weir diverts water to intake from bijayapur khola. Riprap (loose stones) were also provided in the weir. Height of about 2.5m to 3m above the riverbed, it is designed as a run off river type diversion structure. It is seen to be kept at shortest span of river cross section while also considering geological conditions.

The divide walls are provided to prevent cross current i.e. for one directional flow.

Guide walls are provided to channelize river comfortably.

2. Undersluice

Under sluice gates are provided to flush the sediments downstream of river. It regulates 20% of the flood discharge. It controls sediment entry at intake by allowing it to pass downstream.

Two bays with gates are provided, one of which is always functioning. During dry period the gates are closed to supply required water to intake.

3. Intake

The intake structure is located on the left bank of the Bijaypur Khola directly adjacent to the diversion weir. The intake is a side intake type designed to function over free-flow conditions. It includes manual coarse trash racks having 3 orifice at the entrance to block large floating debris (80 % clogging). There are two divide walls provided at the intake and intake gate few meters from intake.

The intake is located at the concave bank to avoid bed load.

4. Intake Canal/ Culvert

About 60 m RCC rectangular closed canal with a design discharge capacity of 8.3 m³ /s connects intake to the gravel trap. Gate operating system is provided to connect water from three sections to single connecting canal.

5. Gravel Trap

A gravel trap is a structure provided after the intake to remove coarse sediment such as gravel, pebbles, and small boulders from the flow before it enters the headrace. It is a widened section where water velocity is reduced for heavy particles to settle.

A RCC structure of about (7m * 3.5 m) removes gravel and debris from water flow. A 242m RCC closed canal connects gravel trap to the desander (approach canal). Flushing mechanism is provided for extracting the trapped gravel.

6. Spillway

To discharge the excess water, overflow spillway of ogee shaped has been provided at downstream of settling basin. When water flows over the crest of spillway, it must always remain in contact with or slide over the surface of spillway. The spilled water is mixed with the river through the channel.

7. Settling Basin

The settling basin of Bijaypur is a dual chamber rectangular structure consisting of enlarged section at entrance to reduce the velocity and hence the turbulence so that the bed load gets settled. The settling basin is of hooper type. The section of the basin becomes converging into similar shape and size of Forebay canal at outlet.

To remove the settled particles flushing arrangement is provided.

8. Fore bay

(*headrace canal/ tunnel is not provided in this hydropower project due to availability of enough head after the level of settling basin according to that topography)

A forebay is a temporary water storage regulating reservoir provided at upstream of penstock to act as small balancing reservoir. In Bijaypur, fore bay has been provided just downstream of settling basin. A spillway has also been provided at side of fore bay to spill the excess water. The water spilled is mixed into the river by dissipating its energy at stilling basin. Flushing gate has also been provided to flush the sediments of fore bay.

9. Penstock

Penstock is a pipe that supplies water from forebay to turbines. Penstock carries water to turbines with least possible loss of head. Penstock pipe starts after a forebay to the power house.

A 634m long, 2.5m diameter welded ribbed penstock pipe carries water from forebay to power house in bijayapur-I hydro plant. Anchor blocks are also provided at every sharp horizontal and vertical bends along the penstock pipe to resist transient pressure developed by water hammer, and expansion and contraction due to variation in temperature. The penstock is bifurcated at the tail to provide water to two turbines.

10. Powerhouse

A structure for housing all essential power generating equipment such as turbine, generator, switchboard, etc. In Bijaypur hydropower station of 23m* 11m , two turbine-generator units are provided. Water from 2 penstock pipe hit turbine. Power house contained service bay, machine bay and control room.

Control room is provided to manually control the power generation. The type of turbine used is Francis turbine. It is reaction turbine and is used for medium head hydropower project. In runner water enters radially and leaves axially so it is also called “mixed flow turbine”. Major part of available water energy is converted into kinetic energy at the entrance to runner and a considerable remaining part is also utilized as pressure energy. Both kinetic and pressure energy are directly converted into mechanical energy by turbine runner before generating electrical energy.

Step up transformer is used to step the produced voltage of 11KV to 33KV.

Gantry Beams are provided for heavy lifting and precision handling of equipment.

11. Tailrace canal

The tailrace is constructed as a cut and cover Reinforcement Cement Concrete (RCC) structure providing a durable and controllable outlet for water after it passes through the turbines. It extends 302m ensuring efficient conveyance of the discharged water back into the river channel.

12. Transmission

The bijayapur-I small hydropower project transmits electricity at 33 KV. The generated power is transmitted via 4 km, 33 KV single circuit line to lekhnath substation, where it connects to NEA 132 KV grid.

13. Access road

A 2 km access road has been constructed, providing direct vehicular access to the powerhouse.

3. Present Scenario

The hydropower sector in Nepal is currently progressing well, with projects like the Bijaypur-I Small Hydropower Project contributing significantly to the national grid. This development has brought several positive impacts, particularly for the local community. The project has an installed capacity of 4.5 MW, with a minimum generation of 2.11 MW during the dry season and a maximum of 4.5 MW in the wet season. The electricity is supplied to the Lekhnath area, highlighting the project's direct benefit to the local population.

4. Discussion and Findings

Our visit to the Bijaypur-I Small Hydropower Project provided a crucial opportunity to connect theoretical knowledge with practical application. We learned that the plant's design, from the diversion weir to the Horizontal-Axis Francis turbines, is well-suited for its medium-head, run-of-river scheme. The cascade system with the downstream Bijaypur-II plant is a particularly noteworthy finding, as it demonstrates an efficient and innovative approach to maximizing water resource utilization. The project successfully fulfills its primary function, contributing 2.11 MW to 4.5 MW to the national grid depending on the season.

However, we identified several operational challenges. The most significant issue is the vulnerability to seasonal river flow, which directly impacts power generation. The plant's operational efficiency is further hindered by observed issues within the turbine section, which, according to a seven-year analysis, are a primary cause of performance impacts. We also noted a lack of consistent maintenance and a pressing need for better management of sediment and plastic trash, which can cause significant damage to machinery and reduce efficiency.

5. Recommendations

Based on discussions and findings, the following recommendations were made to improve the project's sustainability and efficiency:

- **Improve Sediment and Trash Management:** Regular and rigorous removal of sediment and plastic trash is essential. The workers should be provided with better training and tools to effectively manage these issues, which will protect the turbines and increase the plant's lifespan.
- **Enhance Equipment Maintenance:** A structured and regular maintenance schedule for all machinery, particularly the turbines and generators, should be implemented to prevent breakdowns and ensure optimal performance.
- **Study Environmental Impact:** Further study is needed on the effects of the hydropower project on the local aquatic ecosystem. This can help in developing mitigation strategies to minimize environmental damage and maintain river health.

6. Conclusion

The Bijaypur-I Small Hydropower Project is a testament to the potential of run-of-river schemes in Nepal. The field visit allowed us to gain a practical understanding of its components, from the headworks to the powerhouse, and to appreciate the engineering behind its operation. While the project faces challenges related to seasonal flow and maintenance, it remains a vital source of clean energy and a positive influence on the local community. By addressing these issues through improved operational practices and continued investment, such small hydropower projects can continue to play a crucial role in resolving Nepal's energy crisis and fostering sustainable development.

Photos



Figure 2. Stone Lined Weir



Figure 3 Side Intake



Figure 4. Course Trash Racks at Intake



Figure 5 Gravel Trap



Figure 6. Settling Basin



Figure 7. Francis Turbine, Generator, Flywheel (from left)



Figure 8. Generation Control Panel



Figure 9. Gantry Beam inside Powerhouse



Figure 10. Powerhouse



Figure 11. Interconnection @132 kV Lekhnath SS

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